

Email Marketing Campaign Analysis and Click Prediction

1. Problem Statement

The project is not just a coding task but a business problem. The objective of this case study is to evaluate the effectiveness of an email marketing campaign and understand the factors that drive user engagement. In simple words, the company wants to know, “How can we send better marketing emails so that more users click on them?”, based on various email and user-related attributes.

The case study also aims to provide insights that can help improve email marketing campaigns and increase user engagement.

2. Dataset Overview

Three datasets were provided for this analysis:

email_table.csv

This dataset contains information about all emails sent to users. The available attributes include:

- email_id
- email_text (Long or Short)
- email_version (Personalized or Generic)
- hour (Time when email was sent)
- weekday
- user_country
- user_past_purchases

email_opened_table.csv

This dataset contains the email IDs of users who opened the email.

link_clicked_table.csv

This dataset contains the email IDs of users who clicked the link present in the email.

Together, these datasets help analyze user engagement and build a predictive model.

3. Approach

The overall approach was divided into three stages:

Stage 1: Data Loading and Calculations

The datasets were loaded using Pandas DataFrames. Initial inspection of the datasets was performed to understand their structure and verify data quality.

The following checks were performed:

- Data structure inspection
- Missing value analysis
- Verification of column information

After understanding the data, campaign metrics were calculated:

- Open Rate
- Click Through Rate (CTR)

A target variable named “clicked” was also created for machine learning purposes, where:

- 1 = user clicked the link
- 0 = user did not click the link

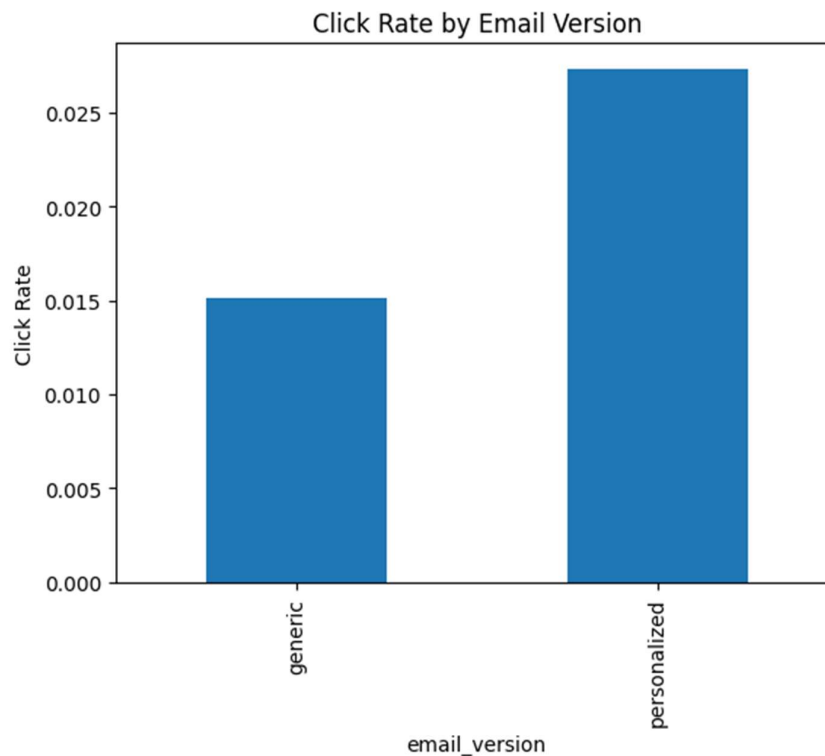
Stage 2: Exploratory Data Analysis and Visualization

Exploratory Data Analysis (EDA) was performed to understand user engagement patterns and identify factors that influence email click behaviour. The objective of this analysis was to explore relationships between different email and user-related attributes and their impact on the Click Through Rate (CTR).

The following comparisons were performed:

- **Personalized vs Generic Emails**

The click rates of personalized and generic emails were compared to determine whether personalization improves user engagement.

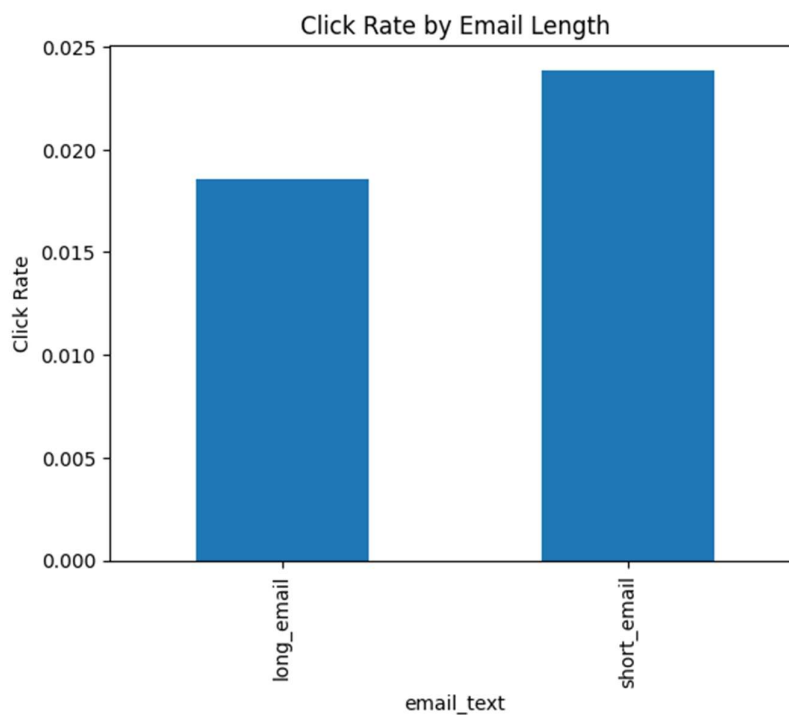


Observation

The result indicates variations in engagement across different days of the week and help identify the days associated with higher click activity.

- **Long vs Short Emails**

The click rates of long and short emails were analyzed to understand the effect of email length on user engagement.

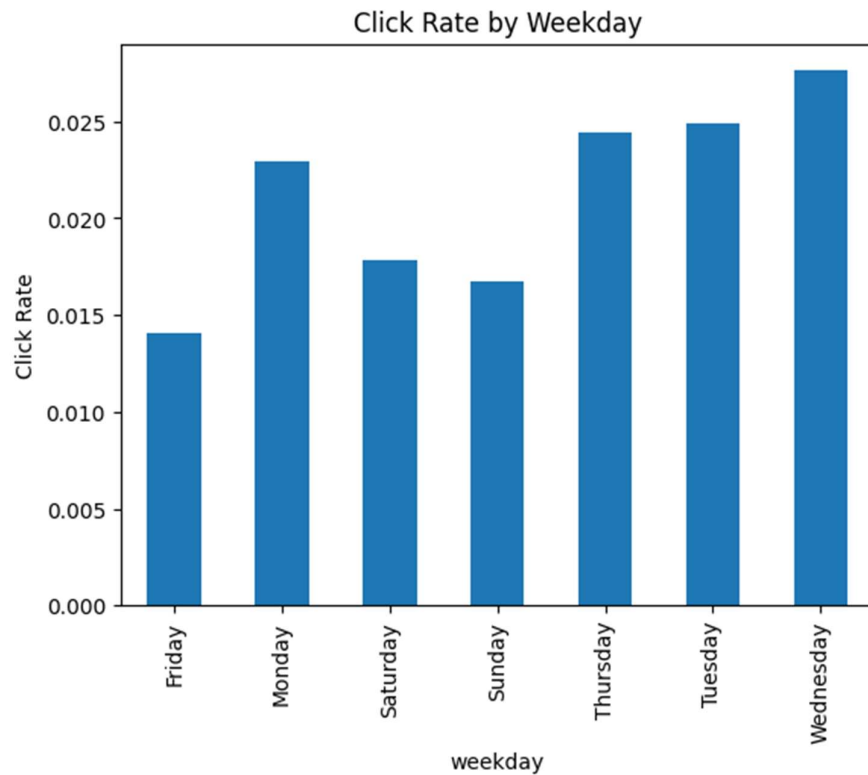


Observation

The analysis highlights which email format performs better in terms of attracting user clicks and engagement.

- **Weekday Analysis**

User click behaviour was analyzed across different weekdays to identify the most effective day for sending marketing emails.

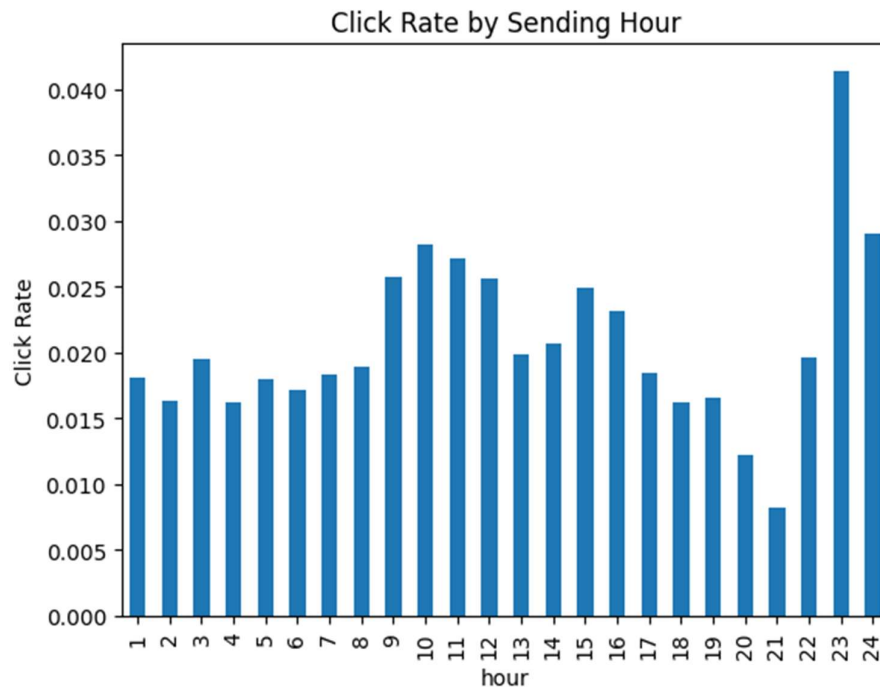


Observation

The result indicates variations in engagement across different days of the weeks and help identify the days associated with higher click activity.

- **Hour-wise Analysis**

Click rates were analyzed based on the hour at which emails were sent.

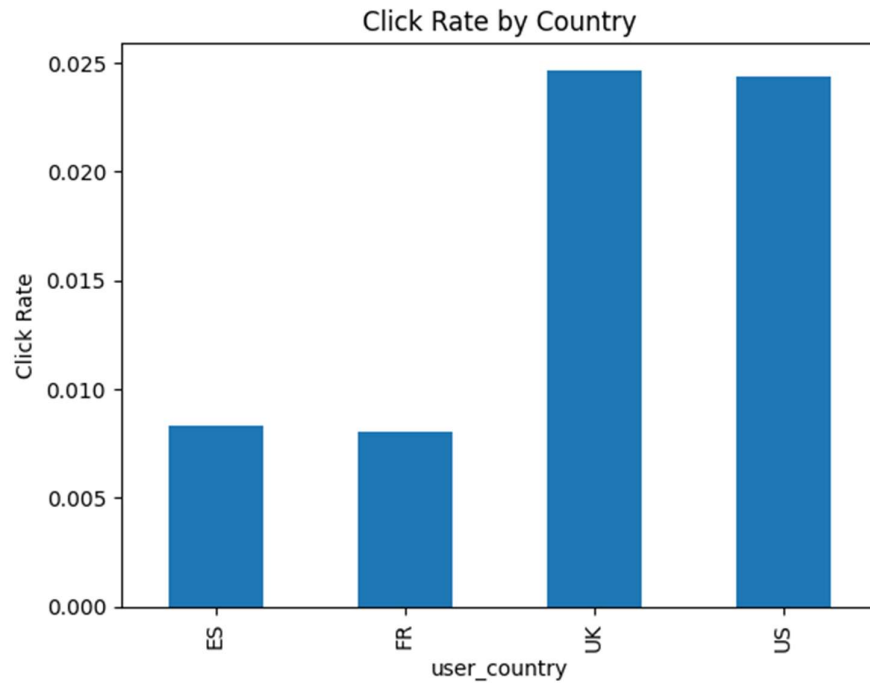


Observation

The analysis helps identify the time periods during which users are most likely to engage with marketing emails.

- **Country-wise Analysis**

User engagement was compared across different countries.

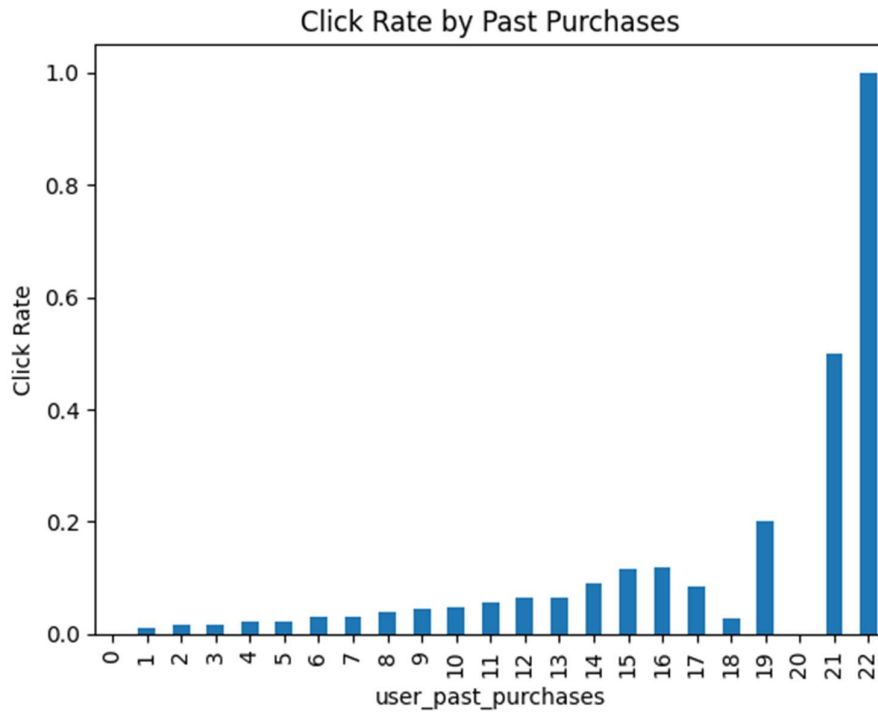


Observation

The analysis reveals geographical differences in user behaviour and helps understand how engagement varies across regions.

- **User Past Purchase Behaviour**

The relationship between previous purchase activity and email engagement was examined.



Observation

User with different purchase histories may exhibit different level of engagement. This analysis helps determine whether past purchasing behaviour influences the likelihood of clicking marketing emails.

Summary of EDA

The exploratory analysis was conducted to identify patterns in user engagement and understand the factors associated with higher click rates. These insights provide valuable business understanding and serve as a foundation for the machine learning model developed in the later stages of the project.

Stage 3: Machine Learning

After understanding the data and user behaviour patterns, a machine learning model was developed to predict whether a user is likely to click on an email link.

The workflow included:

- Feature preparation
- Target variable preparation
- Train-test split
- Model training
- Model evaluation
- Feature importance analysis

Model Selection

We selected **Random Forest Classifier** because it:

- Performs well on structured tabular datasets
- Handles multiple features effectively
- Reduces overfitting through ensemble learning
- Provides feature importance information
- Produces reliable and interpretable results

4. Calculations

- **Open Rate**

Open Rate measures the percentage of users who opened the email.

Formula:

$$\text{Open Rate} = (\text{Opened Emails} / \text{Total Emails Sent}) \times 100$$

- **Click Through Rate (CTR)**

CTR measures the percentage of users who clicked the link present in the email.

Formula:

$$\text{CTR} = (\text{Clicked Links} / \text{Total Emails Sent}) \times 100$$

These metrics provide an overall understanding of campaign effectiveness.

5. Visual Analysis

The visual analysis was performed to identify factors associated with higher engagement.

The analysis focused on:

- Comparing click rates between personalized and generic emails.
- Comparing click rates between long and short emails.
- Identifying the most effective weekdays.
- Identifying the most effective sending hours.
- Comparing user engagement across countries.
- Comparing click rates by past purchases.

These visualizations helped reveal behavioural patterns and business insights that may not be immediately visible through raw data alone.

6. Machine Learning Approach

A Random Forest Classifier was used for clicked prediction.

Feature Selection

The following features were used as model inputs:

- email_text
- email_version
- hour
- weekday
- user_country
- user_past_purchases

The email_id column was excluded because it is only an identifier and does not contain predictive information.

Target Variable

clicked

- 1 = Clicked
- 0 = Not Clicked

Train-Test Split

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

The training data was used to teach the model patterns, while the testing data was used to evaluate performance on unseen observations.

7. Results

Campaign Performance Metrics

The effectiveness of the email marketing campaign was evaluated using Open Rate and Click Through Rate (CTR).

Metric	Value
Open Rate	10.35%
Click Through Rate (CTR)	2.12%

These metrics provide an overall measure of user engagement with the email campaign.

Machine Learning Model Performance

A Random Forest Classifier was trained to predict whether a user would click on an email link.

Metric	Value
Model Accuracy	0.97455

The achieved accuracy indicates the model's ability to correctly classify user click behavior based on the available email and user-related attributes.

Key Findings

The exploratory analysis and machine learning model revealed several important insights:

- Personalized emails achieved a higher click rate than generic emails.
- Email content characteristics influenced user engagement.
- User behaviour varied across different weekdays and sending hours.
- User engagement differed across countries.
- Previous purchase activity showed a relationship with click behaviour.
- The machine learning model successfully identified patterns associated with user engagement.

Feature Important Analysis

Feature importance scores obtained from the Random Forest model helped identify the variables that contributed most to predicting user clicks.

	Feature	Importance
0	hour	0.459203
1	user_past_purchases	0.371352
2	email_text_short_email	0.032536
3	email_version_personalized	0.020092
11	user_country_UK	0.016125

	Feature	Importance
4	weekday_Monday	0.015294
8	weekday_Tuesday	0.014424
7	weekday_Thursday	0.013662
5	weekday_Saturday	0.013204
12	user_country_US	0.012943

The most influential features were:

1. **Hour (0.4592)** – The time at which the email was sent had the strongest influence on click behaviour.
2. **User Past Purchases (0.3714)** – A user's previous purchase activity was highly predictive of whether they would click an email link.
3. **Short Email Content (0.0325)** – Email length played a role in user engagement, with short emails showing a measurable impact on click behaviour.
4. **Personalized Email Version (0.0201)** – Personalization contributed positively to predicting user engagement.
5. **User Country (UK) (0.0161)** – Geographic location also influenced click behaviour, although to a lesser extent than the top features.

These features had the strongest impact on the model's prediction of whether a user would click on an email link.

8. Conclusion

This case study focused on analyzing the performance of an email marketing campaign and understanding the factors that influence user engagement. The analysis began with calculating key campaign metrics such as Open Rate and Click Through Rate (CTR), followed by exploratory data analysis to identify patterns in user behaviour and email performance.

The EDA revealed how factors such as email type, personalization, sending time, weekday, user country and past purchase behaviour influence the likelihood of a user clicking on an email link. These insights provide a better understanding of customer engagement and can help optimize future marketing campaigns.

A Random Forest Classifier was then developed to predict whether a user is likely to click on an email link. The model successfully learned patterns from the available data and demonstrated the potential of machine learning in supporting marketing decisions.

Overall, this project demonstrates how data analysis and machine learning can be combined to generate actionable business insights, improve campaign effectiveness and support more targeted and data-driven email marketing strategies.